

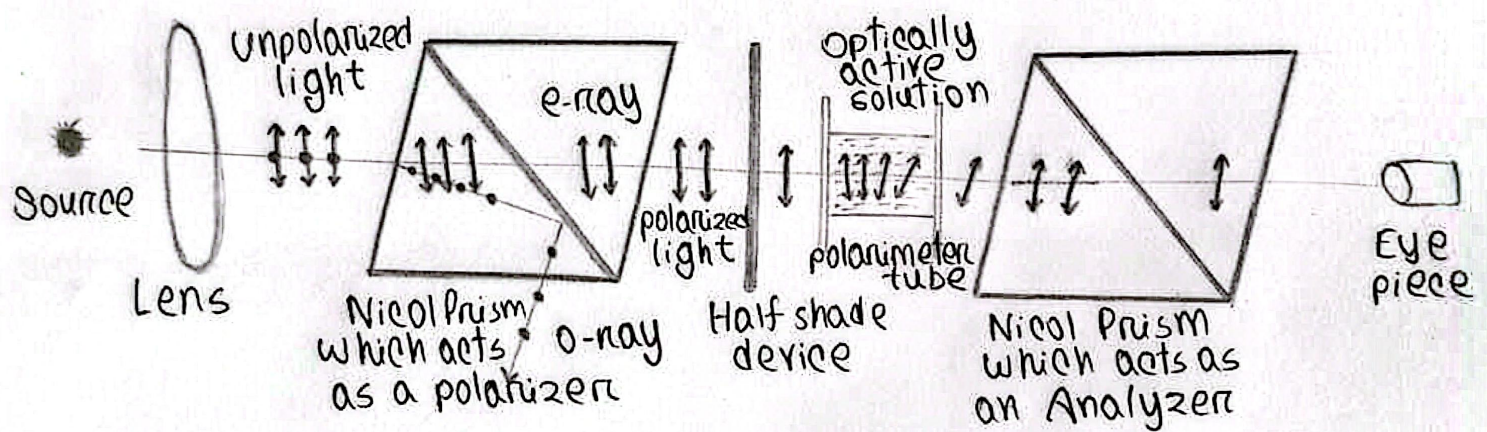


Figure 1: Laurent's Half Shade Polarimeter

THEORY ->>>

EXPT. NO. 01

DATE: 23.11.2025

EXPERIMENT OF Determination of the specific rotation of sugar solution by means of polarimeter**Theory:**

The specific rotation of a solution is a characteristics physical property of a given optically active molecules. If the specific rotation of a compound is positive, the compound is said to be dextrorotatory and if negative, it is referred as levorotatory. The substances that rotates the plane of vibration to the right are called right handed or dextrorotatory and the substances that rotate the plane of vibration to the left are called left handed or levorotatory. The angle of rotation produced to the plane of vibration by an optically active substance in solution is proportional to,

- (i) The thickness of the solution
- (ii) The concentration of the solution or density of active substance in the solution
- (iii) The nature of the substance

Thus, we can write,

Here,

$$\theta \propto lc$$

$$\Rightarrow \theta = S \cdot lc$$

$$\Rightarrow S = \frac{\theta}{lc}$$

If l is in cm, then,

$$S = \frac{100\theta}{lc}$$

 θ = Angle of rotation l = Length of solution (in dm) $c = \frac{m}{V}$ = Concentration of solution ($\frac{g}{cm^3}$) m = mass of sugar (g) dissolved in water V = Volume of sugar solution S = Specific rotation

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In our experiment, usually we measure l in cm and the concentration of the sugar solution is not unit. In that case,

Specific rotation = $\frac{\text{Rotation produced by 1 dm length of sol}^n}{\text{Density of the solution in gm/cm}^3}$

$$= \frac{\theta}{\left(\frac{l}{10}\right)c}$$

$$= \frac{10\theta}{lc}$$

Now, if a graph is plotted with θ against c , then the graph will be a straight line. The polarimeter is thus calibrated

Required Apparatus:

- (i) Polarimeter
- (ii) Balance
- (iii) Measuring cylinder
- (iv) Sugar
- (v) Beaker
- (vi) Thermometer
- (vii) Water
- (viii) Source of light



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Procedure:

- (i) A clean, dry polarimeter tube was selected, and its length were measured in cm using a ruler
- (ii) 3 sugar solutions with concentration of 0.24, 0.18 and 0.12 gm/cm³ were prepared by accurately weighing the required amount of sugar and dissolving it in a measured volume of distilled water
- (iii) The polarimeter was switched on and allowed to stabilize for a few minutes
- (iv) The initial reading of the polarimeter was taken with the empty tube to establish the base line
- (v) The tube was then filled with the first sugar solution, ensuring that no air bubbles were present and the ends were properly sealed. Then the tube was placed inside the polarimeter
- (vi) The angle of rotation produced by the solution was observed and recorded
- (vii) Steps (v) and (vi) were repeated for the remaining two sugar solutions of different concentrations

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(viii) The specific rotation for each solution was calculated using the formula: $S = \frac{100\theta}{lc}$; where,

θ = observed rotation

l = length of tube (in cm)

c = concentration (in g/cm³)

(ix) The average of the three specific rotation (S_1, S_2, S_3) for different concentration was determined as S

(x) A graph of angle of rotation (θ) vs concentration (c) was plotted using the corresponding points and an another average value of specific rotation (S') was obtained from the slope ($m = \frac{\theta}{c}$).

(xi) Then this graphical average specific rotation (S') was then compared with the calculated average specific rotation (S) for verification

FIGURE NO.

calculations:

From Data Table 1,

$$S_1 = 72.7^\circ$$

$$S_2 = 71.99^\circ$$

$$S_3 = 68.45^\circ$$

$$S = \frac{S_1 + S_2 + S_3}{3}$$
$$= 71.05^\circ$$

From graph plotting,

$$m = \frac{AB}{BC} = \frac{4.8}{0.068} = 70.59$$

$$S' = \frac{10\theta}{LC}$$

$$= \frac{10}{l} \times m$$

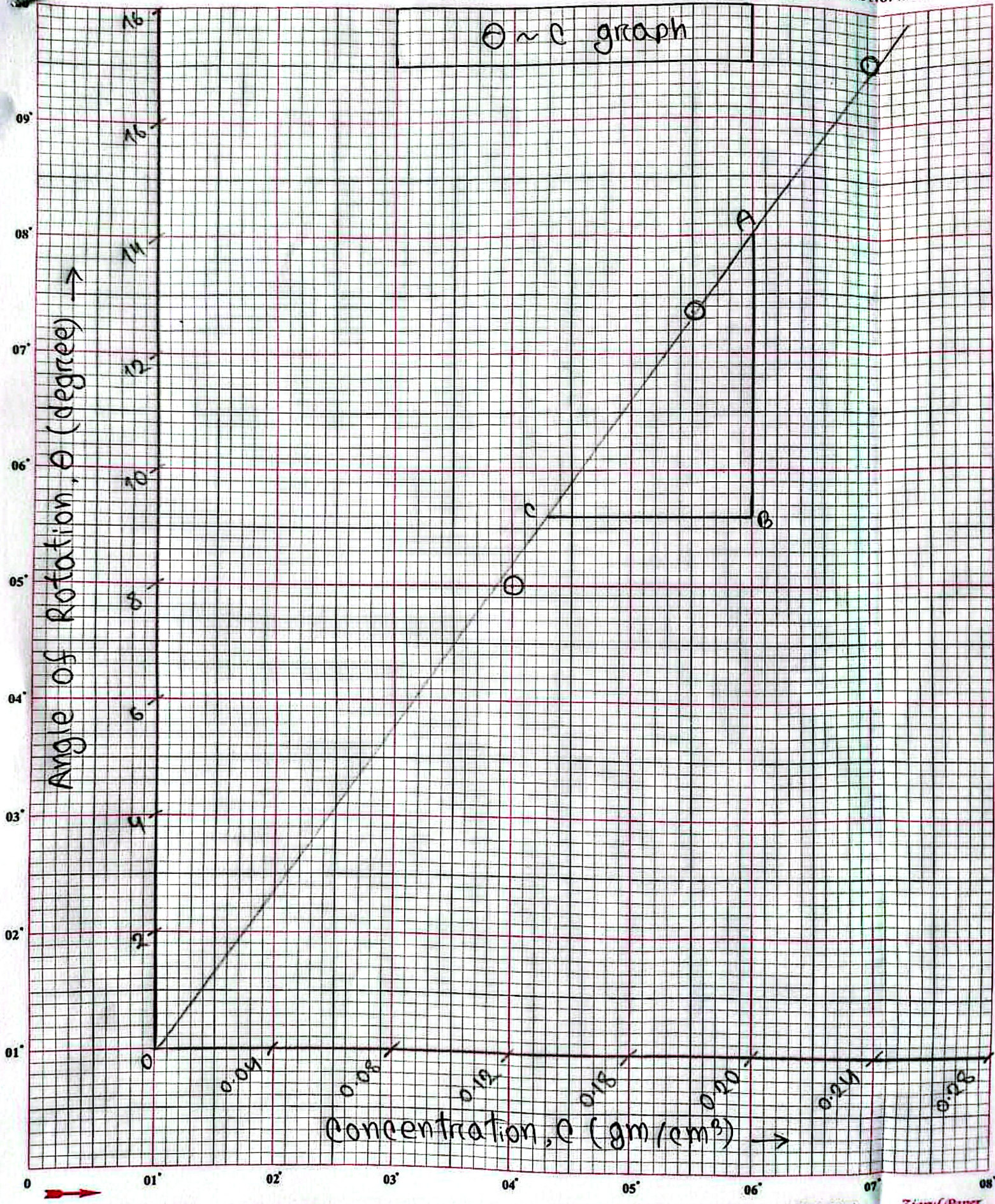
$$= \frac{10}{9.8} \times 70.59$$
$$= 72.03^\circ$$

[GRAPH PAPER]

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$\theta \sim c$ graph



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Experimental Data:

Table 1: Data for determining angular rotation

S. No.	Strength of the solution (%)	Concentration of sugar soln (gm/cm ³)	Scale	Reading with water, P (deg)				Reading with solution, Q (deg)				Angular (Q-P) Rotation (deg)	Mean Angular Rotation (deg)	Specific Rotation (S) (deg)
				MSR (deg)	VSD (div)	VSR (deg)	TR (deg)	MSR (deg)	VSD (div)	VSR (deg)	TR (deg)			
1	24	0.24	A	3	10	0.5	3.5	20	12	0.6	20.6	17.1	17.1	72.7
			B	3	10	0.5	3.5	20	12	0.6	20.6	17.1		
2	18	0.18	A	3	10	0.5	3.5	16	4	0.2	16.2	12.7	12.7	71.99
			B	3	10	0.5	3.5	16	4	0.2	16.2	12.7		
3	12	0.12	A	3	10	0.5	3.5	11	11	0.55	11.55	8.05	8.05	68.45
			B	3	10	0.5	3.5	11	11	0.55	11.55	8.05		

Results:

$$S = 71.05^\circ \text{ (from calculation)}$$

$$S' = 72.03^\circ \text{ (from graph)}$$

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Discussion:

In this experiment, the specific rotation of sugar solution (S) was determined using a polarimeter. By measuring the angle of rotation produced when polarized light passed through the solution. The specific rotation was calculated using the formula $S = \frac{100\theta}{lc}$. Here, we got two sets of specific rotation. $S = 71.05^\circ$ (from calc) and $S' = 72.03^\circ$ (from graph). This slight variation between the values likely arose from measurement errors and experimental inconsistencies. Greater precision and controlled conditions would yield closer agreement.

Precautions:

- The polarimeter tube was well cleaned
- Water used was dust free
- There were no air bubble inside the tube
- Readings were taken when halves of the field of view becomes equally illuminated